

DS 4420: Machine Learning and Data Mining 2

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Electricity Price Prediction

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Introduction

Locational Marginal Prices, or LMPs, represent the price of supplying the next megawatt of electricity onto the grid at a specific point. These prices help operators control flows of power, especially when power lines are overloaded or when electricity demand is high. Being able to predict the LMP at a node on the grid is important for electricity traders and utility owners. With the recent news of retaliatory tariffs placed on power imported from Canada [1], predicting LMPs in the northeast United States will become highly significant, as the US relies on Canada's exported power to help with power demand and energy prices will become volatile. In this project, we hope to come up with a methodology to predict LMPs on an hourly basis in face of the potential volatility in prices within the grid.

Literature Review

The Locational Marginal Price consists of three components, the energy, loss, and congestion components [2]. The energy component is the base price of energy at the location and the loss component is the price of supplying an extra bit of power to overcome losses when power flows through transmission lines [2]. Finally, the congestion component reflects the cost of limitations in the transmission network, because power lines have restraints on how much energy can flow over them at a given time [2]. Due to multiple components and the ability for the grid to change quickly, predicting LMPs is extremely complex.

There are multiple methods that groups have tried to predict the LMP. One of these using a Monte Carlo sampling technique [3]. This technique is used to estimate conditional probabilities between possible components such as time and wind power to predict uncertainties that influence LMP. To model these uncertainties, the authors use random walk, and AR models to show variability in the LMP. The Monte Carlo method provides a type of brute force method to find these predictions which can require a high computational cost [3]. This method, while accurate, may not be plausible with its computational methodology. We were curious if we could use a similar model using time series models where our data would be sequential. More specifically, using ARIMA models to see if features like time would be a good predictor for LMPs while not being computationally expensive to use.

However, due to the hourly nature of our data, problems could arise using an ARIMA or other time series models when predicting multiple hours into the future. When forecasting long enough into the future, these models can converge towards the trend or mean in the data. This means the model would be useful only for a few hours into the future, but a utility owner or power trader might want to see up to 48 hours into the future. Previous work has used rolling forecasts with time series models to help overcome this issue [4], [5]. In these projects, the authors made a prediction, then updated and retrained the models when a new actual data point came in to make subsequent predictions. However, in the article "A rolling forecast approach for next six-hour air quality index track" they generated a forecast for the next 6 hours. When the next hour of data comes in, they discard the original 6 hour forecast and a new one is generated.

This approach yielded a mean absolute normalized gross error of 7.7% on average [4]. In comparison, “An Empirical Study on Internet Traffic Prediction Using Statistical Rolling Model” made a prediction for a test data point, then evaluated it against the observed value. Following this, the model is retrained with the actual value and this retrained model is used to make subsequent predictions. The best performing model using this was the SARIMAX with a 6.83% mean absolute percentage error (MAPE). In contrast, the SARIMAX without the rolling approach yielded a 21.58% MAPE, showcasing that the rolling approach improved the model substantially [5]. These approaches are slightly different, but the concept of using a rolling forecast is something that we can incorporate in predicting LMPs to avoid the issue of the predictions converging towards the trend in the data and hopefully improve the accuracy of the model.

LMPs can be extremely volatile, and might not follow seasonal trends due to changing conditions on the grid. Therefore, introducing a multi-layer perceptron (MLP) to predict LMPs might be useful to try to capture non-linear relationships in the data. However, there would be concerns that the data might lose its temporal component of being a timeseries, and certain seasonalities might be lost. Prior work has used convolutional neural networks [6], [7] in addition to other network structures to make predictions with time series data. The authors of “Convolutional neural networks for time series classification” used convolutional and pooling layers followed by a multilayer perceptron to generate features from the raw time series data, and then the MLP used these features to make a classification. The authors described that the CNN framework is powerful because it discovers the structure of the time series data, and doesn’t rely on human-designed features like specifying a lag or period in the data [6]. They found that the CNN achieved the best classification performance compared to other models like a support vector machine [6]. While this model showed strong performance, this was a classification task and predicting LMPs would be regression based. The authors of “A Long Time Series Prediction Model Based on CNN and Linear Combination” used a combined CNN and linear layers to make predictions of road occupancy rates, electricity consumption, weather, and other regression tasks [7]. They explained that they chose a CNN due to its ability to learn nonlinear features in the time series data, capture patterns within the time series, and the ability to combine the convolutional layers with other models. Overall, they found that this model improved accuracy and offered advantages like reduced complexity and faster runtime. Due to the future work in using convolutional and max pooling layers for forecasting with time series data, we believe that a CNN with a dense layer and a linear activation function would be appropriate for our time series data. It might also be beneficial to have the CNN extract relevant patterns in the data due to how volatile the LMPs can be.

ML Methodology and Data Collection

In order to predict LMPs, we decided to use historical LMP values, load forecasts, and day-ahead prices. The reason we chose load or demand forecasts was because this data would be available before the timestamp has occurred. ISO New England (ISONE) publishes the demand

forecasts up to 48 hours in advance, so this data would be readily available if someone wanted to predict an LMP. Additionally, day-ahead prices are published by ISONE 24 hours in advance so this data would also be available to a point that has not occurred yet. We collected our datasets from GridStatus, a data platform with real time and historical power data [8]. We chose one node to focus on to predict LMPs, Bear Swamp Hydroelectric Station (AR.BEARSWMP13.8BSW1P). This is a pumped storage plant in Massachusetts [9], chosen for its relevance because Massachusetts imports power from Canada, and the pumped storage plant is able to mitigate price volatility because it can store energy when prices are low and produce when prices are high unlike a traditional gas or coal plant that is unable to store power. The data series used in this project were the ISO New England (ISONE) LMP hourly data for Bear Swamp [10], ISONE hourly load (demand) forecasts [11], and ISONE day ahead pricing data [12]. There were no missing values in any of these datasets

First we created time series models to predict LMPs, meaning that we only needed the historical LMP data to create lagged features. We plotted the autocorrelation function (ACF) and partial autocorrelation function (PACF) to determine if an Autoregressive (AR) or Moving Average (MA) model would be most appropriate for the LMP data. In Figure a, the ACF shows significant correlation between lags up until lag 40. The PACF shows that the first 20 or so lags reach significance, but not to the magnitude that the ACF does.

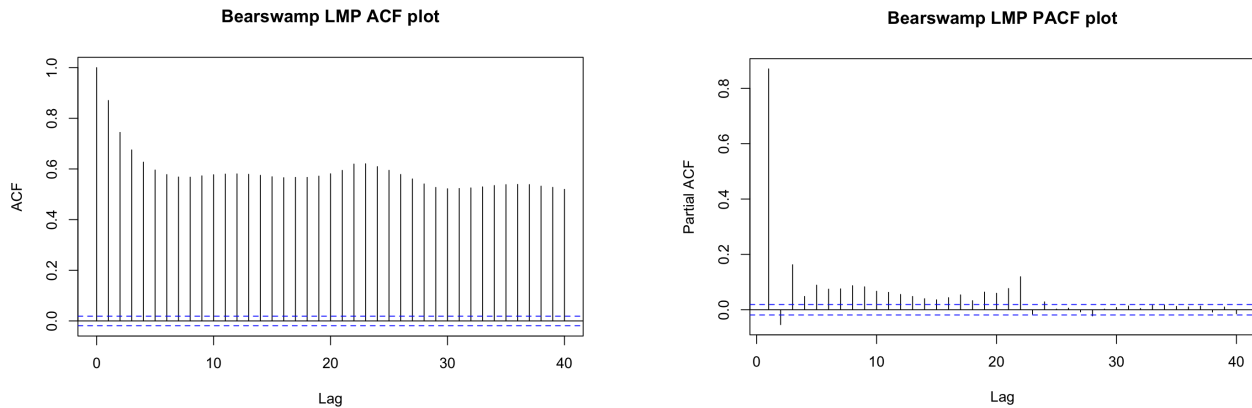


Fig. a. ACF and PACF plots in R from Bear Swamp LMP data

Therefore, we believe that an AR model would be appropriate on these data. Additionally, we performed an Augmented Dickey-Fuller Test (ADF) which checks if the data is stationary. The test revealed that the LMP data is stationary, meaning the data has a stable mean and variance over time, so patterns in the past will be meaningful in predicting the future. An AR model is a time series model where the current value of y (in this case LMP) is predicted by using the lagged values of y from previous timestamps. It is fit using ordinary least squares, giving coefficients for each lag that can be used as a linear model to make predictions. To create the AR model, we determined that a lag of 12 hours would make sense after testing lags of 12, 24, and 48. Lag 12 has information from the past, but isn't too strongly fit to former values due to the volatility of LMPs. To prepare the data for fitting the model, we split into training and test sets.

The first 80% of the data was used to fit the model. We created a matrix of 12 lagged values for X plus a bias column of 1s for the intercept. Then, we used ordinary least squares, taking $(X^T X)^{-1} X^T Y$ on the training dataset to find the coefficients for the model. To see how the model would do on unseen data, we tested it on the test dataset. However, we had to be cognizant of the fact that unseen data would not have 12 lagged values, rather we need to use values we predict as the lags. To do this, the first timestamp in the test set used the last 12 values from the training dataset for the lags. The second timestamp's first lagged value was the predicted value for the timestamp - 1, and the next 11 were from the training set. This kept occurring, where the lagged values became previously predicted values until we reached the end of the test data.

We decided to use a CNN to predict the LMP as it can handle time series data and include multiple features that the time series AR model can't include. We decided to use the features we felt would influence the LMP of the next hour the best. We included month as a feature to see whether a certain time of year would influence the LMP demand, `day_ahead_lmp` which is the company's predicted price for the following day, and the `load_forecast` which is the predicted demand for the whole region of New England will need. We also included a bias term to allow the model to learn better about the relationship. After scaling our data and using a 80% training data set, we started our CNN model.

When implementing the CNN we used the Keras package in python to create our model. We decided to use the 1D convolutional layers and dense layers. We created a simple model as we felt we don't have too complex of data to train the model with. We also ended up using the Adam optimizer because of its good gradient descent capabilities along with it being a good way to learn from less complex data.

ML Results and Interpretation

We experimented with a few lags for the AR model. In the end, we decided that a 12 hour lag would be sufficient, as most power traders are only concerned about the short term, and 24 hours was less accurate across the test set. After fitting the AR(12) model to the training dataset, we had 13 coefficients to make a prediction, with the extra being for the intercept of the model. Figure 2 shows these coefficients, where the index represents the lag + 1 because 1 is the bias term.

	[, 1]		
[1,]	5.260198658	[8,]	0.030395980
[2,]	0.035861377	[9,]	0.025433357
[3,]	0.010830942	[10,]	-0.022088917
[4,]	0.021302151	[11,]	0.120800833
[5,]	0.007479704	[12,]	-0.222903856
[6,]	0.032692451	[13,]	0.844276280
[7,]	-0.023506106		

Fig. 1. Coefficients of the AR(12) model

The intercept coefficient is 5.26, which can be thought of as the baseline price if all lags were 0. Lags 1-9 are all small in magnitude, some being positive and others negative. These have a small effect on the current LMP. Interestingly, lags 10-12 are higher in magnitude. Lag 10 has a positive influence of 0.12, and then lag 11 has a negative influence of -0.22. Finally, lag 12 has the largest coefficient of the lagged terms at 0.84, a strong positive influence. The larger coefficients for the later lags might mean that there is a half-day cycle in LMPs. This could mean that morning and evening has a repeating structure every 12 hours, maybe due to demand patterns or other market behaviors. For example, the LMP at 8AM is strongly positively influenced by the LMP at 8PM the day before, meaning if it was large at 8PM it's probably large now at 8AM. The inverse can be said, if it was low at 8PM, it's probably low now at 8AM. This is an interesting pattern that could be interpreted from the coefficients, but it might not necessarily be because of a causal relationship, but other behavior from demand or broader market behavior that repeats every 12-hours.

The AR(12) model for predicting LMPs showcased a root mean squared error of 85.81 on the test dataset. As the model uses predicted values as inputs for future timestamps, the predictions converged to around \$37 after the first 15 predictions. This is expected behavior of an AR model because without new external information on what the LMPs actually were, the model tends to move towards the long-term trend, which was 37\$. However, when plotting the first 15 predictions of the test set, the model does capture more of the trends within the data, as can be seen in figure 2.

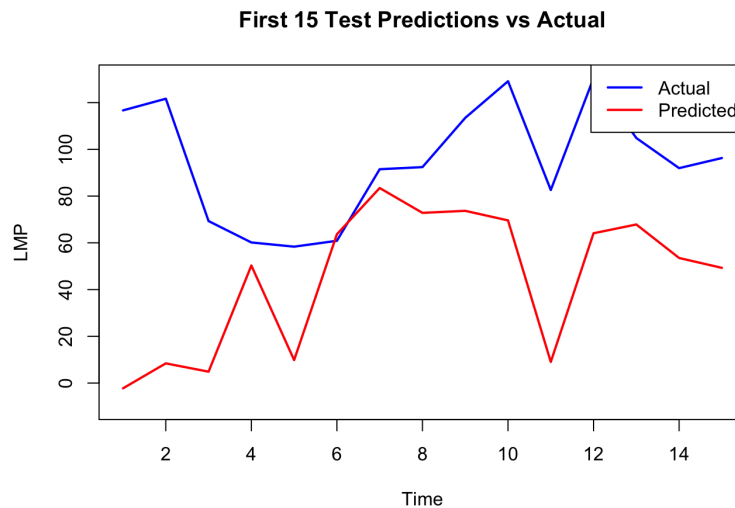


Fig. 2. First 15 predictions vs actuals in the test set for Bearswamp's LMP.

This showcases that when the model has actual data to use in the prediction, it does capture some of the trends within the LMP. For example, it caught the decrease in hour 11, but the magnitude of the LMP was just off. Overall, the AR(12) model unsurprisingly converged to a singular value, but when it can use actual data points it is able to capture some of the patterns in the data. Most people who would be predicting LMPs would be utility owners or traders, who are really

only focused on the next 48 hours. Therefore, this issue might not be as large in practice, where someone could update the data with what the LMP has been over the past couple of hours.

After developing our base Convolutional Neural Network model, we decided we would want to play around with the architecture of the model. Consistently getting random spikes in our loss that would go from 5000 loss to 5 million loss, we would tinker with our model and add more layers like a max pooling layer to hopefully have the model learn more generality about our data. After experimenting with different layers, nodes, and learning rates. We found our best model came from using 50 epochs and a learning rate of 0.015. Using root mean squared error as our metric we found our test set to come out with a 20.74 RMSE loss.

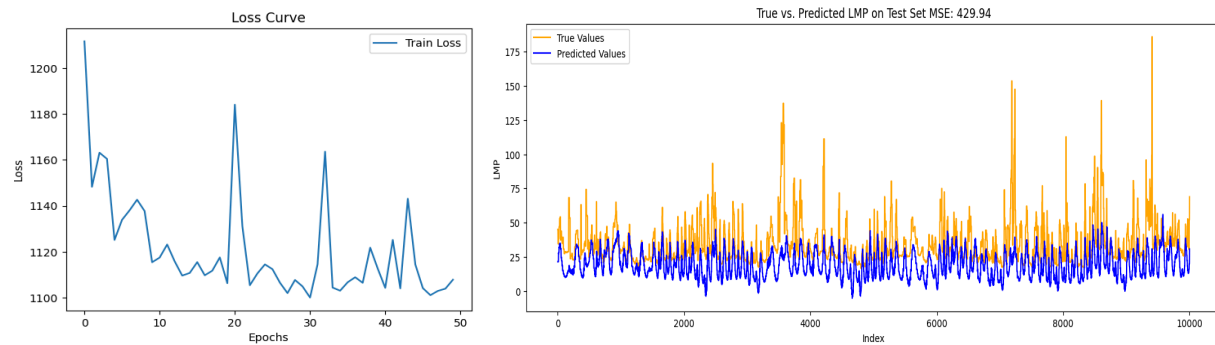


Fig. 3. CNN model's predictions vs actual data values and loss curve for 50 epochs, learning rate of 0.015

When modeling our model's final results for its actual testing data compared to the predicted data, we see the model follows the pattern very well. It understands the LMP's ability to constantly increase and decrease throughout the time running. However, we can see that it never tracks the volatile jumps the LMP may have throughout the time. We see giant spikes throughout that our model will not predict.

With a training loss of around 1000 and a testing loss of 20.74, we decided we may potentially be underfitting or underrepresenting this data. We may be missing out on certain features that would be helpful to add to our training set for our model to predict the true LMPs. To make sure of this, we decided to run this again with a different learning rate and more epochs.

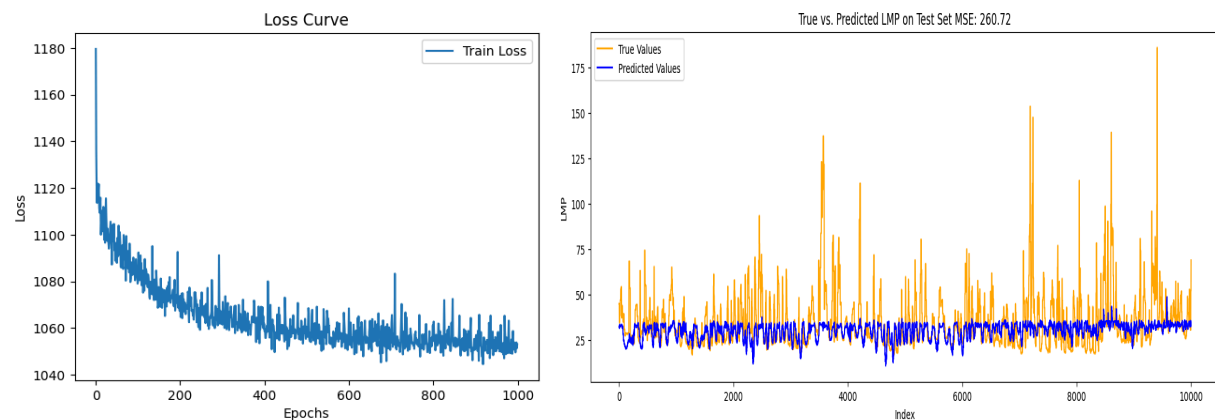


Fig. 4. CNN model's predictions vs actual data values and loss curve for 1000 epochs, learning rate of 0.005

After trying the model again for longer epochs and smaller learning rates. We found the model to have better loss but continue to instead overfit our data and represent the volatile nature of the LMPs even worse. We concluded that our final model should be the one that can represent the normal jumping of highs and lows of LMP data the best, even though the loss may be a bit higher.

Conclusions and Future Work

Both the AR(12) and the CNN with dense layers showcased the difficulty in predicting LMPs due to the volatility in prices. The AR(12) model showcased an RMSE of 85.81 on the test set, and the CNN model performed better with a test set RMSE of 20.74. Both of these models' assumptions were met. In the AR model, the data was stationary according to the Augmented Dickey-Fuller Test. Additionally, the PACF and the ACF revealed that an AR model would be most appropriate for this data. For the CNN, we had the data sorted by timestamp so the convolutional layers were able to find and capture temporal patterns in the time series data. We also ensured that all features that went into the model would be available the day before, so in practice this model would be able to be used to predict a day into the future. Due to all these assumptions being met, we believe that the analysis of the data was appropriate. However, future work could be done to improve these models.

We found the LMPs to be much harder to predict than we originally thought. The LMPs can fluctuate throughout the hour. This gives our models the great task of finding what is the main cause of the spikes in price, which they ultimately did not have much success with. Though, this gives us a ton of room to expand our project with future work, especially with the newly implemented tariffs on Canada which could be an indicator of increased prices that could be added to our model.

An extension of the AR model that could improve accuracy of predictions is the rolling AR model. We discussed the possibilities of a rolling time series model, where the model is retrained when more data comes in in the literature review. Future work could use the AR(12) model implemented here and try to combine it with other methods, like a rolling time series model. This might improve accuracy as the model is able to adjust based on recent trends. However, one of these models might actually do worse, becoming too overfit to the data it's been trained on and not being able to generalize. As discussed, the first 15 predictions on the test set of the AR(12) model were able to capture some of the trends in the LMP. A trader or utility owner would be interested in LMPs in the near future, so this limitation of the model converging to one value might not be as significant if someone is only interested in the next day or so of prices.

As said above, the CNN does a great job of understanding the data's consistent trend but fails to learn its volatile nature. We assume this could be because of our dataset not having the best predictors for these spikes in prices. In the future, we should expand our data set and also

the model. Using more outside factors we did not have access to might explain why the prices jump so high. While having more time to experiment with different activation functions, more network layers, and different learning rates may help as well. We believe it is our data set that is most responsible for the underfitting of the model's predictions.

Overall we felt our project did a good job at predicting these trends but has much room for improvement. Moving forward, using more time series models like rolling AR or ARIMA models would probably be the best predictors for our particular dataset. With better data and increased work on the models we could find a better predictor of the LMPs. Though due to the very volatile and constant changing of the prices, we would not rely on our models to give us solid predictions of what is to come with hourly LMP prices, but they would be good in overall trends and directionality of the prices in general.

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